

Web Crawler Assignment 2 — Report

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Crawl summary: - USERAGENT: IR US26 535578614 79106417 47643578 90303106 - THREADCOUNT: 4 (multithreaded with per-host 500ms politeness) - POLITENESS: 0.5s - Seeds: <https://www.ics.uci.edu>, <https://www.cs.uci.edu>, <https://www.informatics.uci.edu>, <https://www.stat.uci.edu> - Crawl finished cleanly: all 4 workers reported "Frontier is empty" with 0 unhandled exceptions.

Q1. Number of unique pages

5261 unique pages.

Uniqueness is established by the URL after stripping the fragment, per spec.

Q2. Longest page in terms of words

- **URL:** <https://duttgroup.ics.uci.edu/publications>
- **Word count:** 16,436

Words are counted by tokenizing the visible page text (HTML markup excluded, BeautifulSoup `get_text()`) with the regex `[a-zA-Z]{2,}`.

Q3. 50 most common words across all pages

Standard English stopwords are excluded (<https://www.ranks.nl/stopwords>). Tokens are extracted as in Q2.

1. ics, 8629
2. research, 8165

3. data, 7641
4. uci, 7407
5. information, 6410
6. computer, 6040
7. university, 5641
8. us, 5435
9. science, 4874
10. events, 4803
11. students, 4792
12. learning, 4738
13. irvine, 4246
14. systems, 4228
15. search, 4049
16. ramesh, 3929
17. software, 3747
18. news, 3653
19. computing, 3610
20. contact, 3604
21. can, 3478
22. home, 3397
23. pdf, 3333
24. time, 3156
25. student, 3092
26. current, 3078
27. jain, 3030
28. uc, 2963
29. project, 2956
30. new, 2902
31. projects, 2833
32. people, 2821
33. using, 2816
34. machine, 2713
35. support, 2531
36. school, 2526
37. faculty, 2510
38. edu, 2476
39. dataset, 2437
40. paper, 2408
41. past, 2404
42. program, 2401
43. health, 2369
44. based, 2362
45. community, 2343
46. may, 2237
47. graduate, 2187
48. models, 2162
49. week, 2161
50. system, 2142

Q4. Subdomains under uci.edu (100 total)

Alphabetically ordered with unique-page counts.

```
accessibility.ics.uci.edu, 6
acoi.ics.uci.edu, 109
archive-beta.ics.uci.edu, 10
archive.ics.uci.edu, 215
asterix.ics.uci.edu, 7
betapro.proteomics.ics.uci.edu, 1
cdb.ics.uci.edu, 5
cert.ics.uci.edu, 17
checkin.ics.uci.edu, 6
chenli.ics.uci.edu, 10
cloudberry.ics.uci.edu, 5
cml.ics.uci.edu, 37
code.ics.uci.edu, 14
computableplant.ics.uci.edu, 63
containers.ics.uci.edu, 1
courselisting.ics.uci.edu, 2
create.ics.uci.edu, 7
cs.ics.uci.edu, 12
cs.uci.edu, 2
deeprxn.ics.uci.edu, 31
dgillen.ics.uci.edu, 29
ds4all.ics.uci.edu, 9
duttgroup.ics.uci.edu, 106
dynamo.ics.uci.edu, 32
e2.ics.uci.edu, 1
edgelab.ics.uci.edu, 7
elms.ics.uci.edu, 3
emj.ics.uci.edu, 43
evoke.ics.uci.edu, 4
fellowships.ics.uci.edu, 3
flamingo.ics.uci.edu, 13
fr.ics.uci.edu, 3
futurehealth.ics.uci.edu, 179
gitlab.ics.uci.edu, 2
gradinfo.ics.uci.edu, 1
grape.ics.uci.edu, 72
graphics.ics.uci.edu, 1
hack.ics.uci.edu, 1
hobbes.ics.uci.edu, 10
hpi.ics.uci.edu, 5
hub.ics.uci.edu, 4
icde2023.ics.uci.edu, 46
ics.uci.edu, 770
industryshowcase.ics.uci.edu, 24
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informatics.ics.uci.edu, 25
informatics.uci.edu, 7
intranet.ics.uci.edu, 16
ipubmed.ics.uci.edu, 1
isg.ics.uci.edu, 275
jgarcia.ics.uci.edu, 40
julia-hub.ics.uci.edu, 2
luci.ics.uci.edu, 4
mailman.ics.uci.edu, 35
malek.ics.uci.edu, 1
mcs.ics.uci.edu, 12
mdogucu.ics.uci.edu, 1
mds.ics.uci.edu, 9
mhcid.ics.uci.edu, 19
mlphysics.ics.uci.edu, 18
mover.ics.uci.edu, 24
mswe.ics.uci.edu, 10
mupro.proteomics.ics.uci.edu, 3
myip.ics.uci.edu, 1
nalini.ics.uci.edu, 7
netreg.ics.uci.edu, 1
ngs.ics.uci.edu, 737
oai.ics.uci.edu, 6
onboarding.ics.uci.edu, 2
password.ics.uci.edu, 8
pepito.proteomics.ics.uci.edu, 1
psearch.ics.uci.edu, 1
radicle.ics.uci.edu, 6
riscit.ics.uci.edu, 3
scratch.proteomics.ics.uci.edu, 4
seal.ics.uci.edu, 46
selectpro.proteomics.ics.uci.edu, 6
sherlock.ics.uci.edu, 7
signage.ics.uci.edu, 1
speedtest.ics.uci.edu, 1
staging-hub.ics.uci.edu, 1
stairs.ics.uci.edu, 3
stat.ics.uci.edu, 11
stat.uci.edu, 2
summeracademy.ics.uci.edu, 13
support.ics.uci.edu, 4
swiki.ics.uci.edu, 203
tad.ics.uci.edu, 3
tastier.ics.uci.edu, 1
tutoring.ics.uci.edu, 5
unite.ics.uci.edu, 10
vision.ics.uci.edu, 213
wics.ics.uci.edu, 373
wiki.ics.uci.edu, 106

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www-db.ics.uci.edu, 25
www.cs.uci.edu, 5
www.ics.uci.edu, 987
www.informatics.ics.uci.edu, 1
www.informatics.uci.edu, 16
www.stat.uci.edu, 5
xtune.ics.uci.edu, 6
```

Crawler design notes (for grader interview)

What we count as a “page”

A unique URL after fragment-stripping. Even if SimHash near-duplicate detection flags a page as content-similar to one we’ve already analyzed, we still count it toward `unique_pages` (per the assignment’s URL-only uniqueness rule); we just skip its content from word/longest-page analytics so duplicate prose doesn’t bias them.

Politeness

The frontier scheduler enforces a 500ms gap **per host** using a thread-safe `Condition` variable. Workers block in `get_tbd_url` until a host’s gap has elapsed, so politeness is enforced at the scheduling layer rather than via `time.sleep` in workers. We verified zero violations across thousands of requests with 4 worker threads.

Trap detection (`is_valid` / `_is_trap`)

We block URLs matching the following patterns: - More than 10 path segments, or repeating path segments suggesting `/a/b/a/b/...` cycles - Query strings with more than 2 & separators (faceted/calendar combinatorial explosions) - DokuWiki: any query other than `id=` (catches `do=*`, `idx=*`, `rev=*`, etc.) - Trac/MediaWiki revision walkers: `version=`, `rev=`, `revision=`, `oldid=` - Trac action explorers: `action=diff|history|edit|annotate|log|raw|info|render` - Trac timeline scrubbers: query containing both `precision=` and `from=` - WordPress comment-reply / share traps: `repytocom=`, `share=`, `like_comment=`, `unapproved=` - Calendar paginators: `year=`, `month=`, `day=`, `week=`, `tribe-bar-date=`, `eventDisplay=`, `ical=`, `outlook-ical=` - Apache `mod_autoindex` sort variants (`?C=N;0=A` etc.) - Wiki attachment paths (`/raw-attachment/`, `/attachment/wiki/`) - CSS skin variants (`skin=` query param) - Login/logout/auth endpoints (`do=login|logout|register`) - GitLab/Gitweb explorers (`/-/(commits|tree|blob|blame|raw|compare)/`) - Bare blog/category paginators (`/page/N`) - Excluded subdomains: `dale-cooper-v0.ics.uci.edu` (php-fpm status pages), `helpdesk.ics.uci.edu` (IT ticket system, contains PII)

Low-information detection

- Skip pages where `len(tokens) > 5000` and unique-word ratio is `< 0.1` (auto-generated/repeated content)
- Skip non-HTML responses (Content-Type filtering)
- Skip responses larger than 5MB
- SimHash near-duplicate detection: 64-bit fingerprint, blocked when Hamming distance ≤ 3 from any previously-seen page

Resilience

- HTTP retries with exponential backoff (1s, 2s, 4s, 8s) up to 5 attempts before giving up on a URL — handles transient cache-server connection drops without crashing
- Atomic JSON state writes via temp+rename, flushed every 25 pages and on `atexit`
- `try/except` around URL parsing so malformed `<a href>` tags (e.g. literal `[YOUR_IP]` placeholder text from a Trac wiki page) don't crash the worker thread
- `try/finally` in worker's main loop guarantees `mark_url_complete` even if scraping raises